

Regional production structure and fiscal spillovers

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Motivation

- EU spends big to facilitate regional growth - from 2014-2020 352€ billion, or 37% of the common budget.
- Mostly targeted at economically weak regions
- Direct effects and total effects from spillovers are endogenous
- Important for policy and theory in terms of (local) fiscal multipliers, spillover channels, IRF dynamics, etc.

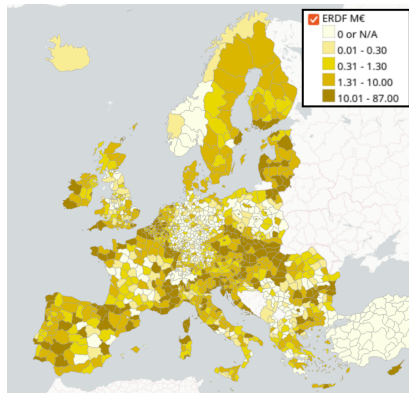


Figure: EU ERDF funding in millions from 2014-2020

EU Cohesion policy at a glance

- Subsidies are paid out from three funds - ESF, ERDF and CF - with a focus on different projects
- National envelopes are negotiated every programming period of 6-7 years - 2007-2013, 2014-2020, 2021-2027
- Allocated sums depend on overall budget but largely follow the "Berlin-rule" - GDP or GNI relative to EU average
- Countries designate individual focus through "operational programs" subject to commission approval
- Allows to negotiate for idiosyncratic circumstances and shift preferences between funds

EU cohesion policy effects

- Credible reduced-form identification so far largely exploits Berlin-rule assignment discontinuities (Becker, Egger, and von Ehrlich 2010; Daminger, Huber, and Nowotny 2026)
- Resulting RDD designs identify local direct effects, but are not well suited to estimating spillovers or total effects
- Evidence from other approaches suggests the total effect is much larger than the direct impact (Fidrmuc, Hulényi, and Zajkowska 2019)
- Existing evidence on spillovers primarily comes from spatial or structural modelling frameworks (Brandsma et al. 2014)

Local Fiscal Multipliers

- Estimates are often with federal US military spending and shift-share IVs (Nakamura and Steinsson 2014)
- Interpreted as relative regional response within a monetary union - a local GE effect
- Actual multipliers are difficult to estimate from these because of interregional trade, migration and other GE adjustments (Guren et al. 2020)
- Reduced-form spillover estimates are a way to approximate the propagation effects via (Auerbach, Gorodnichenko, and Murphy 2020)

Research Gap

- Most research on EU cohesion policy with credible identification is for local treatment effects
 - Rich literature on fiscal multipliers, but very dominant on US military spending
- => Research design for cohesion policy effects that combines credible reduced form identification with rich enough spillover modelling

Main Analysis

The units of analysis are region-sectors, put together in a panel structure:

$$y_{i,t} = \alpha_i + \alpha_{c(i),t} + \beta x_{i,t} + \sum_q \gamma_q (Hx)_{q,i,t} + \delta_{\tau(t)} + \varepsilon_{i,t}$$

This equation is indexed for i at the region-sector level, t being years, $\tau(t)$ the year of a programming period and $c(i)$ being the country:

- **Outcome:** y_{it} region-sectoral GVA or employment
- **Treatment:** x_{it} are the received subsidy payments from EU cohesion funds and β the parameter of main interest
- **Spillovers:** $(Hx)_{it}$ represents the spillovers from treatments to other region-sectors as propagated through a weight matrix H . Specifications are indexed by q .

Shift Share IV as Formula Instruments

Recent research has improved identification on formula instruments with non-random exposure (Borusyak and Hull 2020). This applies to scenarios where shocks follow some formula-based allocation pattern:

- The instrument is a function of exposure shares and shocks

$$Z_{i,p} = f_i(\mathbf{s}_p, \mathbf{g}_p) = \sum_k S_{i,k,p} g_{k,p}$$
- Identification requires $\varepsilon \perp \mathbf{g} | (\mathbf{s}, \mathbf{q})$ - with $\mathbf{q}$ being other observables
- Given that the shocks follow a known conditional distribution $G(g|\mathbf{s}, \mathbf{q})$, one can construct an expected instrument

$$\mu_{i,p} = E(Z_{i,p} | \mathbf{s}_p, \mathbf{q})$$
- Use the recentered instrument for actual exogenous variation

$$\tilde{Z}_{i,p} = Z_{i,p} - \mu_{i,p} = \sum_k S_{i,k,p} \eta_{k,p}$$

The MFF assignment process

Use the Berlin rule and public evidence for negotiations to decompose allocated envelopes into expected components and idiosyncratic deviations from the rule.

$$g_{k,p} = g_{c,f,p} = \Lambda_{c,f,p} F + \eta_{c,f,p}$$

Resulting shocks need to be:

- independent of region-sector-level unobservables
- as good as random under the rule-based component with GDP/GNI relative to EU average, youth and regular unemployment, poverty rates, education levels, land area, infrastructure quality, EU fiscal situation

IV construction

Given the factor model was correctly specified, the residuals can serve as the shifts. Shares can be endogenous, but need to be pre-determined relative to the shocks:

- Revealed absorption: Share of received subsidies relative to the national envelope during the previous period.
- Cofinancing limitation: For each thematic focus and region type, relative limits to EU contributions are set that determine the potential exposure to the funds.

$$\tilde{z}_{i,p} = z_{i,p} - \mu_{i,p} = \sum_k s_{i,k,p} g_{k,p} - \sum_k s_{i,k,p} \Lambda_{k,p} F = \sum_k s_{i,k,p} \eta_{k,p}$$

First-stage Estimation

$$x_{i,t} = \alpha_i + \alpha_{c(i),t} + \lambda_1 \tilde{z}_{i,p(t)} + \phi_1 \mu_{i,p(t)} + \lambda_2 \tilde{z}_{i,p(t)-1} + \phi_2 \mu_{i,p(t)-1} + \delta_{\tau(t)} + u_{i,t}$$

- Effective number of shocks per unit is very low, because regions cluster strictly into countries. Inference at the unit level requires bootstrapping
- Same instrument for each year of the period, $p(t)$ maps years into periods. Fixed effects for year-of-period absorb ramp-up pattern.
- Overlap rule that allows fund use from the previous period requires controlling for the previous period instrument.

Kohesio Microdata

The unit of analysis will be region-sector with NUTS-2 regions and high-level NACE sectors. Outcomes are observed at annual frequency:

- Treatment data must be aggregated from project-level data - available on Kohesio platform
- Requires assignment of NACE sectors to the projects
- Values before 2014 are distributed across different sources and are more labor-intensive

Spillover specifications

$$(Hx)_{i,t} = (Hx)_{s,l,t} = \sum_r \sum_{l'} \omega_{r \rightarrow s} \times d(l', l) \times x_{r,l',t}$$

	No distance	Same location	Contiguous	Distance-weighted
No links		X		X
Direct Forward		X		X
GE Forward				
Direct Backward		X		X
GE Backward				

Table: Possible spillover specifications with primary relevance marked

Spillover instruments

- Construction is straightforward using the recentered instruments and the exposure matrix already in for each specification
- Exclusion restriction states there should be no correlated demand shocks, common factor exposures or otherwise unaccounted elements that drive the dynamics
- Reasonably credible for direct linkages (essentially an SLX structure), but much more difficult for matrices that represent the GE responses
- Different matrices hold different interpretations - labour market transfers for geographical linkages, demand-pull effects for backward linkages, etc.

Data

- Geographical distance between regions
- Input-Output matrices at country level can be regionalised and compared to baseline RHOMOLO tables (inter-regional IO tables)

Conclusion & Open Questions

- Exploit the political assignment process behind EU cohesion policy to extract exogenous variation
- Instrument validity will rest on correctly specified factor model
- Credible instrument for direct effects allows to estimate spillover effects
- Estimates will inform relevant macro debates about fiscal multipliers and the effects of institutional quality on regional growth

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